Philipp Nazari

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Education

MsC **ETH Zurich**, Mathematics — Sept 2023 – Feb 2025

- Focus Areas: Machine Learning, Data Science
- **Research Projects:** Encoder Regularization in Autoencoders, Entropy Aware Message Passing in Graph Neural Networks
- **Thesis:** Why do Neural Networks generalize? Characterizing the probability distribution over the decision boundary realized by a randomly initialized neural network.

BsC **Heidelberg University**, Mathematics — Oct 2019 – July 2023

- Focus Areas: Differential Geometry, Analysis, Machine Learning

BsC **University of Bergen**, Mathematics — Aug 2021 – Jan 2022

- Exchange Semester at the University of Bergen in Norway
- Focus Areas: Algebraic Topology, Differential Geometry

BsC **Heidelberg University**, Physics — Oct 2019 – Aug 2022

- Focus Areas: Theoretical Physics, Machine Learning

Experience

University of Heidelberg, Research Assistant — Heidelberg, Germany, Aug 2022 – Mar 2023

- Worked in the research group of Prof. Dr. Fred Hamprecht, Heidelberg Collaboratory for Image Processing (HCI)

Publications

Geometric Autoencoders – What You See is What You Decode — July 2023

Philipp Nazari, Sebastian Damrich, Fred Hamprecht

proceedings.mlr.press/v202/nazari23a.html (International Conference on Machine Learning)

Research Projects

Geometric Encoder Regularization in Autoencoders — Juli 2024

- Semester Project at ETH with ***Prof. Thomas Hofmann***. Paper and code
- Transferred decoder-based geometric regularization techniques to the encoder

Entropy Aware Message Passing in Graph Neural Networks — Mai 2024

- Project spun out of the Deep Learning course at ETH by ***Prof. Thomas Hofmann***. Paper and code
- Proposed a message passing scheme for graph neural networks that takes into account the entropy of the hidden representations